

INFUSION F&O OPTIONS DASHBOARD

Complete Feature Explanation and Working Summary - v5.0.0 Safety Cockpit

This document explains what the Infusion dashboard currently covers, how each module works, how data moves through the system, and how you should use it for options-first paper trading validation before any future live broker execution.

Item	Current status
Dashboard version	v5.0.0-safety-cockpit
Phase	10/10 complete; pending phases 0
Execution mode	Paper-first only. Live Upstox order placement intentionally disabled.
Primary broker/data source	Upstox token and Upstox market/option-chain data
Trading focus	Indian F&O stocks and stock options, mainly CE/PE buying
Risk baseline requested	Capital Rs 50,000; daily max loss Rs 2,500; preferred start risk 1% per trade

1. The Purpose of Infusion

Infusion is built as an options-first F&O stock screener and trade-decision cockpit. The goal is not just to show price data. The goal is to convince you why a stock is moving, why it may continue or fail, where the important trigger levels are, and whether the option contract is realistic enough to consider.

The system has been remodeled around a practical workflow: scan all F&O stocks, rank candidates, confirm setup quality, check option-chain reality, log paper trades, stage a safe broker-style ticket, and review outcomes. This gives us a learning loop before live automation.

Main idea

- Find directional opportunities early using trend, VWAP, volume, MTF, strength, and conviction logic.
- Avoid theoretical signals that cannot be traded due to spread, premium, lot size, token issues, or data freshness.
- Make every decision auditable using Journal, Execution Staging, Safety Cockpit, and Analytics.
- Keep live order placement locked off until the paper-trade statistics prove the system is stable.

2. High-Level Data Flow

The dashboard is a browser UI, but the intelligence comes from multiple local services connected by Redis, Postgres, and Upstox data. The API service exposes the dashboard endpoints.

Stage	What happens	Output used by dashboard
Upstox auth	Access token is saved and monitored. Expiry is decoded from JWT where possible.	Token status, expiry, recheck trigger, disconnected/expired warnings.
Ingestion	Live market ticks are streamed or fetched and stored in Redis hot keys/streams.	LTP, change %, volume, tick freshness, index prices.
Normalizer	Raw ticks are normalized into consistent symbol-level data.	Clean tick state for scanner/chart/market panels.
Feature engine	Computes VWAP, EMA, RSI, MACD, ATR, Bollinger/compression, relative volume, candle/pattern features.	Scanner intelligence, strength, trend, setup levels.
Scanner	Ranks F&O stocks and builds trade direction, MTF dots, triggers, entry/SL/targets, blockers.	F&O screener rows, selected setup row, active/pre-breakout watch.
API	Combines ticks, features, option chain, news, risk, journal, execution, safety.	All dashboard widgets and tabs.
Dashboard UI	Presents scanner, sectors, options, news, journal, execution and safety in a trading workflow.	Decision cockpit for paper trading.

3. Top Dashboard Areas

Area	What it shows	Why it matters
Header/top strip	NIFTY, BANKNIFTY, selected stock, market regime, A/D, breadth, leader/laggard, session, live status.	Gives immediate market context before stock selection.
Risk console	Manual capital, max daily loss, risk/trade, daily guard, paper-first mode.	Prevents emotional over-sizing and creates a consistent baseline for staged tickets.
Sector strength / heatmap	All sectors with strength score and directional movement.	Helps avoid trading against broad sector pressure.
Breakout + Option command strip	Pre-breakout watch and option basis pinned above screener.	Keeps important setups visible without scrolling.
F&O screener	Main sortable/filterable stock table with fixed symbol/sector and many technical columns.	Primary hunting ground for CE/PE candidates.
Trade workbench tabs	Stock Detail, Triggers, News, Journal, Execution, Safety, Alerts, Diagnostics.	Keeps deeper analysis separated from the main scanner.
Popup chart	Opens selected stock chart in an overlay and TradingView link.	Fast visual confirmation without losing scanner context.

4. F&O Screener - What It Covers

The screener is the core decision table. It is designed to sort the strongest and highest-conviction candidates to the top, while also showing why a row is strong or blocked.

Column/feature	Meaning
Symbol, LTP, Prev Close +/-	Price, previous close, points difference and percentage move.
Change %, RVol, Volume	Momentum and participation. Relative volume is important for option buying.
Sector	Sector classification used with sector strength map.
Rank	Composite smart ranking using strength, conviction, MTF and tradability.
Trend	BUY, SELL, HOLD style direction based on VWAP/EMA/MACD/RSI/ATR logic.
MTF dots	1M, 5M, 15M, 1H, 4H, 1D directional dots. Green/Red/Yellow indicate bullish/bearish/mixed.
MTF Score / Source	Shows how aligned the timeframes are and whether score is historical/proxy/limited.
Strength	Setup quality based on compression, volume, trend and breakout readiness.
Bias	BUY CE, BUY PE, HOLD, or AVOID recommendation for options direction.
Conviction	Options-first confidence score combining scanner and tradability logic.
Why / Block	Human-readable reasons and anti-chase rejection blockers.
Entry, SL, T1/T2	Underlying-level trade map for paper planning.
Bull Above / Bear Below	Trigger levels where the stock turns positive or negative.

Scanner logic ingredients

- VWAP relationship: CE preference generally needs price above VWAP; PE preference generally needs price below VWAP.
- EMA trend stack: price vs EMA 5/9/20/50 for short-term and broader trend direction.
- MACD and RSI: confirms momentum and avoids some overextended conditions.
- ATR and trailing levels: helps build stop-loss, targets, and chase detection.
- Bollinger/compression and NR patterns: identifies coiled/pre-breakout candidates.
- Relative volume and spread: filters whether the move has participation and liquidity.
- Sector strength: upgrades or downgrades stocks depending on sector alignment.
- Anti-chase filters: flags VWAP stretch, large candle, extreme RSI and poor entry conditions.

5. Option Chain and Contract Realism

A stock moving is not enough for options trading. The selected option contract must also be realistic. Infusion therefore checks Upstox option-chain data where available and creates an option readiness view.

Option check	What it does
Bias selection	Chooses CE or PE from scanner direction and underlying movement.
Expiry resolution	Uses Upstox option contract/chain data to find usable expiry.
Near-ATM scoring	Scores strikes near the underlying spot, avoiding far illiquid contracts.
Premium check	Blocks contracts with missing/zero live premium.
Spread check	Wide bid/ask spread is downgraded or blocked.
OI/OI change	Open interest and OI change support liquidity/participation scoring.
IV and delta/moneyness	Avoids unsuitable strike distance and abnormal option structure.
Expiry runway	Avoids contracts that are too close or too far for the intended trade style.
Lot size	Passed into execution staging so quantity and risk are realistic.

Option execution status

Status	Meaning
TRADE_READY	Contract quality has passed major gates. Still paper-only; use TradingView confirmation.
WAIT_CONTRACT	Underlying may be interesting, but option contract quality is not enough yet.
AVOID_CONTRACT	Contract has hard blockers or poor tradability.
CHAIN_PENDING	Underlying scanner is available, but option-chain data is not ready.

6. Pre-Breakout Watch

The pre-breakout watch is a shortlist of stocks that are compressing, coiled, accumulating, or nearing important levels. It is shown above the scanner so you do not miss developing setups.

- Shows symbols that are building toward a possible move before a full active signal.
- Uses readiness/strength style scoring rather than only live signal count.
- Useful for preparing TradingView charts before the actual trigger happens.
- Still requires confirmation from scanner direction, MTF, VWAP, volume and option-chain gates.

7. Selected Stock Detail and Popup Chart

When a row is selected, the dashboard shows inline row details and selected setup information rather than hiding everything in a separate dialog. A popup chart can be opened for quick visual review, and the TradingView button opens the selected symbol in TradingView for your own visual confirmation.

Feature	Use
Inline selected row details	Shows entry, SL, targets, triggers, MTF, why/block directly in the screener row.
Selected trade plan	Shows a compact decision card with option conviction, risk budget and gate checklist.
Popup chart	Quick chart overlay in dashboard; closes when another stock is selected or manually closed.
TradingView button	Opens TradingView so you can visually confirm the structure and levels. No TradingView API is required for this basic workflow.

8. News Edge

News Edge adds context from public news sources. It does not replace price action, but it flags whether headlines appear bullish, bearish, risky or neutral.

News stance	Meaning
BULLISH	Positive headline/event bias that may support CE-side confidence.
BEARISH	Negative headline/event bias that may support PE-side caution or direction.
EVENT_RISK	Risky event words detected; reduce size or wait for stronger price confirmation.
NEUTRAL	No strong directional evidence from headlines.
NO_NEWS	No useful recent news found from current sources.

News Edge currently uses public/news keyword logic. A stronger ChatGPT-powered event classifier can be added later for deeper reasoning, but the current module already helps prevent blind trading during event risk.

9. Telegram Alerts

Telegram remains the notification layer. The goal is that each meaningful alert should be understandable without opening the dashboard immediately.

- Signal direction and grade/score.
- Entry, stop-loss, target levels and R:R.
- Reason for signal and blockers/suppression if applicable.
- Option contract status and suggested contract when available.
- Daily recap with precision, grade, session, sector, regime and suppression details.

If Telegram shows timeout or mismatch with dashboard, the exact timestamp and symbol should be captured for debugging.

10. Journal - Evidence Capture Layer

The Journal is where a scanner idea becomes a recorded paper trade idea. This is the memory of the system. Without journaling, we cannot honestly know whether the scanner is improving.

Captured item	Why it matters
Symbol, sector, decision	Stores what was considered and whether it was CE/PE/HOLD/WAIT.
LTP, entry, SL, T1/T2	Records the trade map at the time of decision.
Strength, conviction, MTF score	Lets us later compare high-score vs low-score outcomes.
Option contract summary	Stores contract quality, premium, spread, OI, IV, expiry, instrument key and blockers.
News edge	Stores whether news was supportive, risky, or irrelevant.
Why/block reasons	Preserves the reasoning and anti-chase warnings visible at decision time.
Outcome marking	Win, Loss or Skip is logged after the paper trade result.

Journal workflow

- Select stock in screener.
- Check inline setup, selected trade plan and TradingView visual confirmation.
- Click Log Paper Trade.
- Open Journal tab and verify all evidence is captured.
- After movement, mark Win, Loss or Skip.

11. Execution Staging - Broker-Style Ticket Preview

Execution Staging converts a journal setup into a broker-style paper ticket. It still does not place orders. It tells you if the idea is practical enough to be staged.

Ticket field	How it works
Contract/instrument key	Uses Upstox option contract details from the journal snapshot.
Side/product/type	Currently BUY, INTRADAY, LIMIT style preview for option buying.
Limit price	Uses live option premium if available.
Quantity/lots	Calculated from lot size and risk budget. If budget cannot afford one lot risk, it blocks.
Option SL	Uses a conservative premium-risk estimate; currently 28% premium risk in staging.
Estimated max loss	Quantity multiplied by option premium risk.
Underlying trigger	Uses Bull Above for CE or Bear Below for PE direction.
Blockers	Missing premium, missing lot size, wide spread, missing instrument key, hard blockers, no direction or too-small risk budget.

This layer is intentionally strict. It is better to block a ticket than show a fantasy quantity or trade an illiquid option.

12. Safety Cockpit

The Safety Cockpit answers the most important question before trading: Is the system safe enough to even consider paper trading now?

Gate	Meaning
Paper-first lock	Confirms live orders are disabled.
Kill switch	Manual dashboard block. If ON, the system verdict becomes blocked.
Upstox token	Checks token validity and expiry.

Gate	Meaning
Market data	Checks tick count and freshness, adjusted by market session.
Morning risk	Warns if capital/risk settings were not updated today.
Services	Checks ingestion, normalizer, feature-engine and API health.
F&O universe	Confirms symbols are loaded.

Safety verdicts

Verdict	Meaning
PAPER_READY	No major blockers. Paper trade only, still confirm visually.
WATCH_READY	No hard block, but warnings exist such as risk not updated or non-market hours.
BLOCKED	Do not trade. Clear the blocker first.

13. Risk Console

The Risk Console is manual by design. Every morning you should enter your capital and max daily loss. This avoids hidden assumptions and keeps the dashboard aligned with your actual trading capital.

Setting	Current guidance
Capital	User baseline: Rs 50,000. Update manually every morning.
Daily max loss	User baseline: Rs 2,500 or lower in volatile days.
Risk/trade	Recommended paper validation start: 1%. Avoid jumping to 5% until results prove stability.
Loss-day cap	If 2 or 3 losses happen, stop fresh trades and switch to observation mode.
Execution mode	Paper-first only; auto orders disabled.

14. Analytics, Optimizer and Forward Proof

Analytics helps measure whether the system is actually improving. Earlier phases added precision analytics, optimizer and forward proof checks.

Analytics feature	Purpose
Signal precision	Tracks target hits, stops, expiries and overall outcome quality.
By grade/sector/session/regime	Shows where signals work or fail.
Suppression stats	Shows why signals were blocked.
80% precision optimizer	Looks at archived signal outcomes and suggests stricter guard settings.
Forward proof	Separates future corrected results from old contaminated/buggy data.

Important: historic optimization is useful, but the next serious evidence must come from new paper trades after the corrected logic. That is why Journal and Execution Staging were added.

15. Token and Connectivity Handling

The system now depends fully on Upstox. The dashboard includes token status and a local token dialog/recheck workflow so token expiry is easier to recover from.

Feature	How it works
Token status endpoint	Shows token state, expiry time, ingestion state, tick count and tick age.
Dashboard token dialog	Allows pasting a fresh access token directly from the dashboard.
Force recheck	Writes a Redis signal asking ingestion to recheck token/auth state.
Safety integration	Safety Cockpit blocks or warns if token is missing/expired.

16. Manual Triggers

Manual triggers are for your idea: if this stock crosses this spot price, I want to watch it green or red. They help track levels before scanner generates a full signal.

- Create price levels for selected stocks.
- Use positive-above and negative-below logic from scanner as reference.
- Use triggers as watch alerts, not as automatic entries.
- Confirm with MTF, VWAP, volume and option contract before journaling.

17. How A Trade Idea Should Move Through the System

Step	Action	Decision rule
1	Scanner ranks all F&O stocks.	Look only at strong rows, not random movement.
2	Check sector and market health.	Prefer trades aligned with broad context.
3	Select stock and read inline setup.	Understand entry, SL, target, trigger and blockers.
4	Check Option Basis.	Do not proceed if chain/contract is weak or pending.
5	Open TradingView.	Visual confirmation is required before journaling.
6	Log Paper Trade.	Only log if idea is clear and low-risk.
7	Stage Ticket.	If ticket is blocked, do not override casually.
8	Track paper outcome.	Mark Win, Loss or Skip in Journal.
9	Review analytics.	Use outcomes to tune scanner parameters later.

18. What The System Covers Today

- F&O stock universe scanning.
- Live Upstox data integration and token recovery workflow.
- NIFTY/BANKNIFTY and market context at top of dashboard.
- Sector strength and heatmap-style market pulse.
- Options-first scanner with CE/PE/HOLD/AVOID logic.
- MTF dots for 1M, 5M, 15M, 1H, 4H and 1D style alignment.
- VWAP, EMA, MACD, RSI, ATR, volume, compression and candle/pattern logic.
- Anti-chase rejection reasons and blocker explanations.
- Option-chain scoring with spread, premium, OI, IV, strike, expiry and lot-size realism.
- News edge context and event-risk warning.
- Telegram alert and recap layer.

- Manual triggers.
- Paper trade journal.
- Execution ticket staging without live orders.
- Safety cockpit and manual kill switch.
- Analytics, precision optimizer and forward-proof baseline.

19. What Is Not Yet Live

These are intentionally not active or still require future validation:

- No automatic live Upstox order placement from the dashboard.
- No guaranteed 80% live precision claim. The system has guards and optimizer logic, but live paper validation must prove actual performance.
- News intelligence is not yet a full institutional news terminal. It is a useful public-news edge layer.
- Lot-size and option-chain fields should be validated across many F&O stocks during live market.
- The current execution staging is a preview and safety filter, not an order router.

20. Recommended Next Validation Plan

Day/phase	What to do
Day 1	Run dashboard full market day. Do not live trade from it. Log only paper trades.
After Day 1	Send screenshots of Scanner, Journal, Execution, Safety and Telegram issues.
After 30-50 paper trades	Review precision by setup, sector, time, score and blocker type.
After stable results	Tune thresholds: MTF minimum, conviction minimum, spread, RVol, session filters.
Only after strong proof	Consider designing a separate live-order module with explicit confirmations and hard risk locks.

21. Daily Operating Summary

Your daily workflow should be simple: update risk, check Safety, scan F&O rows, confirm with TradingView, log paper trade, stage ticket, mark outcome, review. The dashboard is now structured to support that loop.

The most important habit is not taking every signal. The edge comes from waiting for clean alignment: strong scanner rank, supporting sector, clean MTF, VWAP agreement, good relative volume, no chase blocker, realistic option contract, and safe risk sizing.